

CS/CE 224/272 Object Oriented Programming: Practice Lab

Submission is through Canvas.

P9.18 Write a class `Bug` that models a bug moving along a horizontal line. The bug moves either to the right or left. Initially, the bug moves to the right, but it can turn to change its direction. In each move, its position changes by one unit in the current direction. Provide a constructor

```
Bug(int initial_position)
```

and member functions

```
void turn()
void move()
int get_position()
```

Sample usage:

```
Bug bugsy(10);
bugsy.move(); // Now the position is 11
bugsy.turn();
bugsy.move(); // Now the position is 10
```

Within main, implement the following functionality: Starting with a bug at location 100, make the bug turn randomly based on a random integer (If the random integer is even, the bug turns, else it does not turn). Make the bug move each time whether it turns or does not turn. After 25, 50, 75 and 100 iterations, print the position of the bug.

Naming Convention: Name your program as `p1_<last-5-digits-of-your-id>.cpp`

Engineering P9.23 Define a class `ComboLock` that works like the combination lock in a gym locker, as shown here. The lock is constructed with a combination—three numbers between 0 and 39. The reset function resets the dial so that it points to 0. The `turn_left` and `turn_right` functions turn the dial by a given number of ticks to the left or right. The `open` function attempts to open the lock. The lock opens if the user first turned it right to the first number in the combination, then left to the second, and then right to the third.



```
class ComboLock
{
public:
    ComboLock(int secret1, int secret2, int secret3);
    void reset();
    void turn_left(int ticks);
    void turn_right(int ticks);
    bool open() const;
```

Write a main function that creates a `ComboLock` with 3 secret keys. Then simulate opening of the lock by making various left and right turns. Make sure you print various turn actions taken in the main.

Naming Convention: Name your program as `p2_<last-5-digits-of-your-id>.cpp`

P9.13 Define a class `Country` that stores the name of the country, its population, and its area. Using that class, write a program that reads in a set of countries and prints

- The country with the largest area.
- The country with the largest population.
- The country with the largest population density (people per square kilometer or mile).

Write a main function that tests this functionality. You may test with the following data:

COUNTRY	AREA	POPULATION
China	9,596,961	1,410,000,000
Nigeria	923,768	230,000,000
Germany	357,022	84,000,000
United States	9,833,520	341,000,000
Brazil	8,515,767	218,000,000
Australia	7,692,024	27,000,000

Naming Convention: Name your program as `p3_<last-5-digits-of-your-id>.cpp`
